Time complexity

program → exec. time

how do we calculate? ?

seconds/mins/xyz?

unit operations

3 nested

loops → single
→ nested

x seconds

ops → program
→ efficien ⊗

a+b

a−b

a*b

a/b

a%b

→ input

output → c_1 && c_2

c_2 || c_1 ✓

a=1;

{ a>b

b==c

a<=b }

a++

f---

{ 1 unit

ops }

total measure ← combining

platform → any program

time allotted

heavy

⇓

5s

TLE

1s

program

OS

Kill

Tc1

Tc2

Tc3

Tc4 .....

execution

1s

res 1 → expected output

WA

✓

×

$1s = 10^8$ unit ops.

for(i=1; i<=n; i++){
print(i);
}

input console → n

print(i); → unit ops

unit ops ?

(3n+2)

TC → representations

O( ) → worst case

i=1 → 1

i<=n → n+1

i++ → n

print() → n

i=1 <=n

i=2 <=n

i=3 <=n

⋮

i=n <=n → True

i=n+1 <=n → False

+/O/+
+ +
x x

$O(\cancel{3}n+\cancel{2})$

$\downarrow$

$O(n)$ = unit ops?

how many times my loop is running.

$3n+2$

$\downarrow$

remove consts.

$\times 1$

$3 \cdot n$

$\downarrow$

$O(n)$

{ worst case. }

best avg case

j=1, j=2 . . . j=n

i=1 : 1, 2 . . . . . . $n \to n$

i=2 : 1 . . . . $n \to n$

i=3 : 1 . . . . $n \to n$

⋮

i=n : 1 . . . . $n \to n$

$n * n \to O(n^2)$

```
for( i=1 ; i<=n ; i++) {
    for( j=1 ; j<=n ; j++) {
        print(j) ;
    }
}
```

$i \to n \; \{ \; j \to n$

$O(n*n)$

for(i = 1 ; i <= n ; i++) $\rightarrow n$

print(i);

}

for(i = 1 ; i <= m ; i++) $\rightarrow m$

print(i) ;

n

$+$

m

$O(n+m)$

$2^0 \leftarrow i = 1) * 2$

$2^1 \leftarrow i = 2)$

$2^2 \leftarrow i = 4)$

$2^3 \leftarrow i = 8)$

$2^4 \leftarrow i = 16)$

$1 \dots n$

$i = 2^0$ $\quad i = 2^x$

$2^x <= n$

n → input

```
for(i=1; i<=n; i*=2)
{
    print(i);
}
```

TC

$n = 14$

$i = 1, 2, 4, 8, 16$

$n = 16$

$i = 1, 2, 4, 8, 16, 32$

$\boxed{2^x = n}$

$i = 1, 2, 4, 8 \dots 2^x$ $\quad n$ $\quad 2n$ $\quad 2^{x+1}$

$0 \dots x = x+1 \rightarrow x+1$ iters

$\log_2(a^b)$

$\|$

$b * \log_2(a)$

$\log_a(a) = 1$

$2^x = n$ ✓

$\log_2(2^x) = \log_2(n)$

$\boxed{x \cdot \log_2(2)} = \log_2(n)$

$x \cdot 1 = \log_2(n)$

$x = \log_2(n)$

$\{0 \ldots x\}$

$\boxed{x+1}$

$O(\log_2(n) + 1)$

$\downarrow$

$O(\log_2(n))$

$\frac{n}{2^0} \leftarrow n > 0$

$\frac{n}{2^1} \leftarrow n/2 > 0$ ②

$n/2^2 \leftarrow n/4 > 0$

$n/2^3 \leftarrow n/8 > 0$ ① int div

$n/2^4 \leftarrow n/16 > 0$

$n/2^x = 1$ $n = 1 > 0 \longrightarrow n = 0$ ⊗

n

while $(n > 0)$

$n /= 2;$

$\} \rightarrow$ time complexity

$\frac{n}{2^0} \rightarrow \frac{n}{2} \rightarrow \frac{n}{4} \cdots \frac{n}{2^x} = 1$ 0

$x+1$ iteration

$$\frac{n}{2^x} = 1$$

$0 \cdots x$

$$\frac{2^x}{n} = 1 \Rightarrow 2^x = n$$

$x+1$

$$x = \log_2(n)$$

$$O(\log_2(n)) \leftarrow O(\cancel{\log_2(n)+1})$$

$$\begin{array}{r|l l}
2 & 5 & 1 \\
2 & 2 & 0 \\
2 & 1 & 1 \\
 & 0 &
\end{array}$$

$(101)_2$

$\frac{5}{2} = \frac{2}{2} = 1$ //

$8/2 \to 4/2 \to 2/2 \to 1$ //

$15/2 = 7/2 \to 3/2 \to 1$ //

$\Rightarrow$ (Binary rep. of a decimal value.)

$1/2 = 0$

$$\begin{array}{r|l l}
2 & 8 & 0 \\
2 & 4 & 0 \\
2 & 2 & 0 \\
2 & 1 & 1 \\
 & 0 &
\end{array}$$

$= (1000)_2 = 8$ //

$O(n + \log(n))$!

$n = 10 \rightarrow 10$

3

$n = 100 \quad n \rightarrow 100$

$\log(n) \rightarrow 7$

$\log_2(100)$

$2 * \log_2(10) \simeq 7$

```
for( i=1; i<=n; i++)  → n
    print(i);

for( i=1; i<=n; i*=2)
    prin(i);
```

$\log(n)$

~~log(1)~~

n=1

$\log_2(1)$

0

log(n)

log(n)

n

n

log(n)

O(n)

$n = 10 \rightarrow n = 10$

$n^2 = 100$

$O(n^2 + n)$?

$n^2 \rightarrow 10^8$

$n \rightarrow 10^4$

$n = 100 \rightarrow n = 100$

$n^2 = 10^4$

$n = 1000 \rightarrow n = 1000$

$n^2 = 10^6$

$O(n^2)$

$n^3 + \cancel{n^2} + \cancel{n} + \cancel{\log(n)}$

$O(n^3)$

{TC} → $n = 100$

$n = \underline{\underline{100}} \longrightarrow$ $O(n) \to 100$ — 10^2

$O(n^2) \to 10^4$ — 10^8

$O(n^3) \to 10^6$ — 10^8

$O(n^4) \to 10^8 = = 10^8$

$n = 1000 \to$ $O(n) \to 1000$

$O(n^2) = 10^6$

$O(n^3) \to 10^9$

$10^8 \to 1s$

$10^9 \to \frac{10^9}{10^8} = 1s$

$10^9 \to$

CPT

{overhead}

$\boxed{O(1)}$ constant K

$\frac{n*(n+1)}{2}$

$n = 10^5 \rightarrow O(n) \rightarrow 10^5$

$O(n^2) = 10^{10}$ X

$O(\log_2(n)) =$

$\log_2(10^5)$

$5 * \log_2(10)$

$5 * 3.5 \approx 20$

$n = 10^{18}$

$O(1)$ $O(n)$ X

$O(\log_2(n))$

$18 * \log_2(10)$

$\approx 50 - 60$

$n = 10^8$

$O(n) \rightarrow 10^8$ X

10^8

$O(1)$

$O(\log_2(n))$

$18 * \log_2(18)$

$\approx 50 - 60$ $< 10^8$

$\theta \rightarrow$ avg. complexity

$\Omega \rightarrow$ best time complexity